

Divorce-induced health decline and the impact on mortality risk*

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Abstract

Divorce is linked to elevated mortality. However, the causal mechanisms underlying this relationship are complex and remain insufficiently identified. Health is one such mechanism and is the focus of this paper. Using linked administrative data from the German Pension Insurance Fund (DRV), we examine whether divorce increases the risk of an initial musculoskeletal rehabilitation, and how much of the divorce-mortality gap runs through this channel. Rehabilitation is administered by the DRV and is granted only once a musculoskeletal condition has become chronic and threatens the ability to work. Our outcome measures the crossing of an institutional threshold rather than the onset of disease. We build annual risk sets of individuals experiencing a first (observable) divorce and individuals who have not yet divorced in that year and then use propensity score matching to generate comparable matches. **Results: Divorce effect on cumulative incidence of musculoskeletal rehab, divorce effect on mortality, results from decomposition**

Keywords: divorce, health, rehabilitation, mortality

JEL Classification: H75, I12, J12

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1 Introduction

The link between divorce and mortality has been widely discussed in social epidemiological studies. Register-based studies have used Finnish data (Metsä-Simola and Martikainen, 2013), Norwegian data (Berntsen and Kravdal, 2012) and English/Welsh data (Grundy and Tomassini, 2010). Until recently, Germany has been a difficult setting for this question. Brockmann and Klein (2004) is the only study to the best of our knowledge which explicitly analyzes the link between divorce and mortality although the dataset used in the study is the SOEP¹ which is not an administrative dataset. Since 2011, it is possible to link different administrative datasets from the German Pension Insurance Fund (DRV) using a unique identifier, allowing us to access full income biographies, divorce information², rehabilitation history and death. The availability of this dataset led to the conception of the project *Divorce and Diagnosis (DiDi): Register-based Analysis on the Role of Divorce in Health and Employment Dynamics*³, which constitutes our study.

Our study focuses on the interaction between divorce, health and mortality. While there can be a direct link between divorce and mortality if divorce results in suicide (Kposowa, 2000) or homicide, causal evidence remains scarce when the link is indirect. Hence, we chose health as it is one possible indirect link tying together divorce and mortality. There can, however, still be threats to identification as health is not only a consequence but also a cause of divorce (Torvik et al., 2015). Furthermore, the individuals who choose to divorce may not necessarily be random. Many of the register-based studies make use of hazard models to estimate excess mortality risk after divorce and controls for sociodemographic and socioeconomic variables. For Germany, Brockmann and Klein (2004) account for health by using the self-reported health measure from the German SOEP and control for variables associated with marital status to find that divorced men have 51% higher mortality than married men while there is no significant effect for women. Leopold (2018) presents a very careful multi-outcome treatment of the gendered consequences of divorce over time, but it is not designed to control for the health that occurs before from the health consequences which follow the divorce. In our paper, we control for pre-existing health trends and other factors which occur before the divorce by using propensity score matching to answer the following questions

RQ1: Does divorce raise the risk of musculoskeletal rehabilitation⁴ among divorced and non-divorced individuals?

RQ2: Of the total effect of divorce on mortality, how much operates through an indirect channel (musculoskeletal rehabilitation) and a direct channel?

For context, the DRV only grants rehabilitation once the musculoskeletal condition is chronic to the point that it threatens working ability (Deutsche Rentenversicherung Bund, 2005). We do not argue that divorce causes musculoskeletal disease but rather whether divorce moves individuals who carry a latent vulnerability across the legal

¹German Socioeconomic Panel

²This information is available due to the law of splitting pension points.

³This project is a collaboration between the Federal Institute for Population Research, Hertie School, Charité - Universitätsmedizin Berlin and the DRV.

⁴Musculoskeletal diseases are the most common cause of lost work time in Germany.

threshold where rehabilitation becomes necessary. Please see Section 2.3 for a more detailed discussion.

Results... maybe policy implications?

Section 2 provides a literature review, Section 3 introduces the data and methodology, Section 4 presents the results, Section 5 discusses the implications and Section 6 concludes.

2 Literature review

2.1 Link between divorce and mortality

2.2 Divorce and health

2.3 Divorce and musculoskeletal disease

Rehabilitation system in Germany

Inclusion criteria and ‘exclusion’ criteria for applying for rehab with DRV
https://www.deutsche-rentenversicherung.de/DRV/DE/Reha/Warum-Reha/voraussetzung_ausschlussgruende.html

vague definition of ‘threatening ability to work’, typical duration of medical rehabilitation, cost of rehabilitation, how transitional allowance kicks in when individuals have exhausted the salary receipt from employers.
<https://www.deutsche-rentenversicherung.de/DRV/DE/Reha/Medizinische-Reha/Allgemeine-med-Reha/allgemeine-med-reha.html>

transitional allowance
<https://www.deutsche-rentenversicherung.de/DRV/DE/Reha/Warum-Reha/uebergangsgeld.html>

78 week rule
https://www.gesetze-im-internet.de/sbg_5/__48.html Individuals can only receive pay from their health insurance for a maximum of 78 weeks in a 3-year period

specific prerequisites for musculoskeletal rehab
https://www.deutsche-rentenversicherung.de/SharedDocs/Downloads/DE/Experten/infos_fuer_aerzte/begutachtung/leitlinien_rehabeduerftigkeit_stuetz_bewegungsapparat_kurzfassung_pdf.pdf Basically, it is not sufficient to only have pain. The criteria which need to be fulfilled in order to qualify for musculoskeletal rehab with the DRV is chronic pain syndrome, incapacity to work with a psychosocial dimension, comorbidity and active cooperation of the insured person in the rehabilitation. In terms of musculoskeletal rehabilitation, there needs to be significant restrictions of movement associated with chronic pain of a persistent or recurrent nature, postural deformities, whiplash distortion, muscular atrophy, etc.

2.4 Musculoskeletal disease and death

3 Data and methodology

3.1 Data

The Research Data Center of the DRV allows linkages of different datasets starting from the reporting year 2011 through a unique identifier (Forschungsdatenzentrum Rentenversicherung, 2025). The datasets we use are AKVS⁵, LTVS⁶, RTBN⁷, RTWF⁸, RSD⁹ and VA¹⁰. The time period covered by the data is 2011 to 2023.

The AKVS, LTVS and RTBN are merged by the DRV to form the BASE dataset. The AKVS dataset contains individuals who are still actively contributing to the pension fund hence it generally represents individuals who are employed. The LTVS dataset contains individuals who have paused pension fund contributions but have not yet retired (nor drawn from their pension fund). It may include individuals who are unemployed, undergoing further education, on parental leave, etc. The RTBN dataset contains individuals who are drawing from their pension fund, i.e. retired individuals. The merged dataset, BASE, allows us to map out the full life-course coverage of individuals. If we only considered only RTBN, the sample is biased towards individuals who have contributed and lived long enough to draw a pension, ignoring those individuals who have undergone rehabilitation or died before drawing a pension. Considering only AKVS ignores rehabilitation efforts and gray divorces. To round up the dataset, it is also important to consider LTVS which details what happens between employment spells and retirement.

We then supplement the merged dataset with the RTWF, RSD and the VA. The RTWF details pension terminations due to deaths. If the death information from the merged dataset is inconsistent with the information in the RTWF, the individuals (cases) are dropped {about 127 individuals are dropped}. The RSD dataset details the completed rehabilitations and include information such as the broad diagnosis group, the type of rehabilitation benefits applied for, start and end date of rehabilitation. The RSD dataset also includes individuals with withdrawn, transferred or rejected applications. We only include individuals whose applications were successful and who completed the rehabilitation spell. Furthermore, about 5% of individuals undergo three or more rehabilitation spells per year. We also drop these individuals. We then supplement the merged dataset by including the additional rehabilitation information for the remaining individuals in the RSD dataset. The VA dataset includes information related to divorce and only contains divorced individuals. From the VA, we can obtain the date of divorce, pension supplements and deductions which result from divorce.

⁵Active Insured Persons

⁶Latent Insured Persons

⁷Insured Persons' Pension Portfolio

⁸Termination of Pensions due to Death

⁹Completed Rehabilitation in the Insurance History

¹⁰Pension Equalization Statistics

3.2 Matching

Since divorce is not randomly assigned, divorced and non-divorced individuals may differ systematically before divorce occurs. We therefore use 1:1 matching without replacement to construct a comparison group of individuals who are similar with respect to observed pre-treatment characteristics. The causal interpretation relies on conditional exchangeability (Rosenbaum and Rubin, 1983). This means that, conditional on the observed characteristics used in matching, individuals who divorce and comparable individuals who do not divorce are assumed to have had similar potential health trajectories in the absence of divorce. In other words, it is assumed that there are no remaining unobserved factors that jointly affect the probability of divorce and the subsequent health outcomes. However, the administrative data may not fully capture factors such as relationship conflict or social support.

We define treatment as the first observed divorce from the earliest observed marriage spell. We limit the treatment cohorts to divorces observed between 2012 and 2018 to ensure that each cohort can be observed for at least five years after divorce. The year of divorce is fixed as t_0 . For each treatment year, we form a separate within-year risk set. For treated individuals whose first observed divorce occurs in t_0 , eligible controls are individuals who have not yet experienced a divorce by that year. Controls may divorce later. We do not restrict the control group to individuals who never divorce because this would define control eligibility using information that occurs after t_0 . This means that it is possible that an individual who was a control before their own divorce can be counted as treated afterwards. Individuals must be alive at t_0 and must have an observed record in $t_0 - 1$. We exclude individuals with a musculoskeletal rehabilitation up to and including t_0 . They may previously have received rehabilitation for other conditions. The resulting sample therefore consists of individuals who are at risk of their first observed musculoskeletal rehabilitation after t_0 .

We require treated and control individuals to be matched exactly on sex, five-year age band and whether any rehabilitation has occurred by $t_0 - 1$. Then, we estimate a logistic propensity-score model separately for each treatment year. The propensity score represents the estimated probability of divorce in t_0 , conditional on observed characteristics measured before divorce. The characteristics we include are age, cumulative rehabilitation history, mental-health rehabilitation history, years since the most recent rehabilitation, pension contribution periods,¹¹ earnings, pension income, unsuccessful rehabilitation applications, federal state, broad occupational group, income-source status and marital status at $t_0 - 1$.

We treat missingness in earnings, pension income and contribution periods as potentially informative. In particular, observed earnings with missing pension income generally indicate active employment, whereas observed pension income with missing earnings generally indicates pension receipt. We retain the observed earnings and pension values and include a categorical measure describing the joint observability of the two income sources. We also include separate indicators for missing contribution-period information. This allows the matching model to use both the recorded values and the information contained in their missingness patterns.

Within each treatment year and exact-match group, we use optimal matching. The

¹¹Periods during which pension contributions were recorded.

matching procedure first maximizes the number of treated individuals who can be matched within the caliper and then chooses the combination of pairs that minimizes the total propensity-score distance. We limit matches to a maximum absolute propensity-score difference of 0.01. Matching is conducted without replacement, meaning that each treated individual and each control can appear in only one matched pair. To accommodate the large annual control pools, the propensity-score model is fitted using all treated individuals and, where necessary, a reproducibly selected sample of controls, with the total estimation sample limited to 500,000 observations. The resulting propensity scores are corrected for the control-sampling fraction, and the fitted model is then used to calculate propensity scores for every eligible treated individual and control. The computational limit therefore affects how the propensity model is estimated but does not remove controls from the matching pool.

We assess covariate balance before and after matching using standardized mean differences. Balance is assessed separately for continuous variables, binary variables, each level of categorical variables and indicators of missingness. We present these results in Figure blablabla. An absolute standardized mean difference below 0.10 is treated as evidence of adequate balance on the corresponding observed characteristic. Where residual imbalance above this threshold remains after matching, the affected variables are additionally included in the outcome model. This adjustment is intended to address limited remaining imbalance and is not used as a substitute for unsuccessful matching.

Using the matched sample, we estimate the average treatment effect among the matched treated individuals. We examine first entry into musculoskeletal rehabilitation, time to first musculoskeletal rehabilitation, rehabilitation duration and all-cause mortality after divorce. Musculoskeletal rehabilitation is examined over a five-year follow-up period, with death treated as a competing event. Mortality is followed for up to ten years where permitted by the end of the observation period in 2023. The amount of available mortality follow-up therefore varies across treatment cohorts.

3.3 Identification

For identification, we focus on the average treatment effect among the matched treated individuals.

3.3.1 RQ1: Musculoskeletal rehabilitation

We follow each matched individual from t_0+1 to t_0+5 and estimate a discrete-time hazard model:

$$\begin{aligned}\lambda_{i\tau}^{\text{rehab}} &= \Pr(R_{i\tau} = 1 | \text{no musculoskeletal rehab before } \tau, \text{ alive at } \tau) \\ &= \Lambda_{\text{rehab}}(\alpha_\tau + \beta D_i + \mathbf{x}_i' \gamma)\end{aligned}\tag{1}$$

where $R_{i\tau}$ is a dummy indicating individual i 's first entry into musculoskeletal rehabilitation in time in follow-up year $t_0 + \tau$ where $\tau \in \{1, \dots, 5\}$, α_τ are fixed effects, D_i is a dummy indicating divorce at t_0 and $\mathbf{x}_i$ are other variables measured at $t_0 - 1$ that we

add depending on the specification. We estimate two specifications for Equation 1. In the first specification, we omit $\mathbf{x}_i$ and rely only on matching to deal with selection bias. In the second specification, we include variables whose absolute standardized mean difference after matching exceed 0.10 into $\mathbf{x}_i$. Standard errors are clustered at the level of the matched pair.

The hazard odds ratio from Equation 1 is difficult to interpret on its own as it does not show how frequently rehabilitation occurs, hence we also estimate the cumulative incidence of a first musculoskeletal rehabilitation over the 5-year follow up period using a discrete Aalen-Johansen estimator¹² (Aalen and Johansen, 1978). We report four values at the the 5-year horizon, 1) cumulative rehabilitation risk among treated individuals, 2) cumulative rehabilitation risk among controls, 3) risk difference per 1000 individuals and 4) the corresponding risk ratio. Confidence intervals are obtained by resampling whole matched pairs.

The biggest threat to the conditional exchangeability assumption is unobserved pre-divorce trends, which available variables may not fully capture, especially by only observing the year prior to the divorce. To address this, we first conducted an event study from $t_0 - 3$ to $t_0 + 5$ by taking the difference in values of unsuccessful rehabilitation applications, earnings, pension income, missingness of pension income and prior rehabilitation activity between the treated and control in each matched pair and conducting a Wald test (Wald, 1943) to determine whether the differences across time are significant for each variable. Following divergences in trends prior to t_0 , we repeated the matching exercise with three lags, i.e. $t_0 - 3, t_0 - 2, t_0 - 1$ instead of only $t_0 - 1$. Since we require that individuals have an observed record, the lag-3 specification considers only the time period between 2014 and 2018. We also repeat the lag-1 specification for only 2014 to 2018.

3.3.2 RQ2: Mortality

We follow each individual from $t_0 + 1$ until $t_0 + 10$ or 2023, the last year of the sample, whichever occurs first and estimate a discrete-time hazard model similar to Equation 1:

$$\begin{aligned}\lambda_{i\tau}^{\text{death}} &= \Pr(Y_{i\tau} = 1 | \text{alive at beginning of follow-up year } \tau) \\ &= \Lambda_{\text{rehab}}(\alpha_{\tau} + \beta D_i + \mathbf{x}_i' \gamma)\end{aligned}\tag{2}$$

¹²This estimator recognizes that individuals who die can no longer enter rehabilitation. For example, suppose there are 100 people where 10 enter rehabilitation in $t_0 + 1$, 20 die before rehabilitation and 70 are still alive and have not entered rehabilitation. Suppose in $t_0 + 2$, 7 people out of the remaining 70 enter rehabilitation. If we treat death as censoring rather than as a competing event, the proportion surviving up to $t_0 + 1$ is $1 - \frac{10}{100} = 0.9$, and up to $t_0 + 2$ is $1 - \frac{7}{70} = 0.9$, $\therefore$ the cumulative rehabilitation probability is $1 - (0.9 \times 0.9) = 0.19$, which is an overestimation. With the Aalen-Johansen estimator, we consider the proportion which actually entered rehabilitation by each follow-up year before dying. In $t_0 + 1$, the proportion entering rehabilitation is 0.1, however, after $t_0 + 1$, the proportion remaining alive without having entered rehabilitation is $1 - \frac{10+20}{100} = 0.7$. In $t_0 + 2$, $\frac{7}{70} = 0.1$ is the share of people who enter rehabilitation, but since only 70% of the original population was alive without having entered rehabilitation at the beginning of $t_0 + 2$, the contribution to the incidence is $0.7 \times 0.1 = 0.07$, hence the cumulative rehabilitation probability is 0.17

where $Y_{i\tau}$ is a dummy indicating that individual i dies in follow-up year $t_0 + \tau$, α_τ are time fixed effects, D_i is a dummy indicating divorce at t_0 and $\mathbf{x}_i$ are other variables measured prior to t_0 . Just as in Equation 1, we estimate two specifications for Equation 2 by varying the inclusion of variables in $\mathbf{x}_i$ based on absolute standardized mean differences after amtching. We complement the pooled hazard odds ratio with cumulative mortality risks using the Kaplan-Meier estimator (Kaplan and Meier, 1958). We again report cumulative mortality risk among treated and controls, the risk difference per 1000 individuals and the corresponding risk ratio at the 5-year horizon¹³. We also run the lag-3 specification with more pre-divorce years.

In addition to the straightforward analysis of divorce on mortality, we run a mediation analysis to determine how much of the mortality difference can be attributed to ‘early’ musculoskeletal rehabilitation. We use landmark periods of $L \in \{1, 2\}$. This is to account for individuals who die early without a rehabilitation ever being recorded since this individual would be unmediated. We chose one or two years because a longer mediator window may have the potential to invite health shocks unrelated to divorce to produce musculoskeletal rehabilitation. Furthermore, since our data is only until 2023, a longer mediator window shortens the mortality follow-up window. With $L = 1$, the horizon, H_L is four years and with $L = 2$, it is three years. The mediator for individual i is defined as:

$$M_i(L) = \begin{cases} 1, & \text{if individual } i \text{ enters musculoskeletal rehabilitation} \\ & \text{between } t_0 + 1 \text{ to } t_0 + L, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

We excluded individuals who died before and at $t_0 + L$ and individuals who do not have an annual record in the mediator window. This is to ensure that $M_i(L) = 0$ indicates that the individual was observed throughout the mediator window without entering rehabilitation instead of rehabilitation cannot be observed because the individual died or disappeared. We then estimate the probability of early musculoskeletal rehabilitation:

$$\Pr(M_i(L) = 1 | D_i, \mathbf{z}_i) = \Phi_{\text{rehab}}(\kappa + \delta D_i + \mathbf{z}_i' \eta) \quad (2)$$

where $\mathbf{z}_i$ contains all available pre-treatment matching variables for both lag-1 and lag-3 specifications. Mortality follow-up begins after the mediator window has closed, i.e. at $t_0 + L + 1$. For the post-landmark follow-up year s , we estimate:

$$\begin{aligned} \lambda_{is}(L) &= \Pr(Y_{is} = 1 | \text{alive at the beginning of post-landmark year } s) \\ &= \Phi_{\text{death}}(\alpha_s + \theta_D D_i + \theta_M M_i(L) + \theta_{DM} D_i M_i(L) + \mathbf{z}_i' \rho) \end{aligned} \quad (3)$$

where Y_{is} is a dummy indicating that individual i dies in follow-up year $t_0 + L + s$ where $s \in \{1, \dots, H_L\}$. As before α_s indicates time fixed effects and D_i is a dummy

¹³The hazard model uses up to ten years of follow-up although later follow-up years are available only for earlier divorce cohorts.

indicating divorce. $D_i M_i(L)$ is an interaction term between divorce and early musculoskeletal rehabilitation. After predicting the probability of death in the follow-up year $t_0 + L + s$, we calculate the cumulative mortality risk:

$$\hat{R}_i(d, m) = 1 - \prod_{s=1}^{H_L} [1 - \hat{\lambda}_{is}(d, m)] \quad (4)$$

4 Results

4.1 Descriptive summary

How many individuals are in the 20% dataset? How many person years? How many people have undergone divorce in each year? We limit matching to years 2012 to 2018 and 2014 to 2018. Who do we filter out? How many people undergo medical rehabilitation? What are the top rehabilitation codes and cases?

Of the N_T eligible treated individuals, N_M were successfully matched, corresponding to a matching rate of (X%). Because matching is 1:1 and without replacement, the matched sample contains N_M treated individuals and N_M unique controls. We additionally report the number of unmatched treated individuals, matching rates by treatment year, the distribution of propensity scores, and the mean and maximum propensity-score distances between matched individuals.

figure to show the event study coefficients and maybe Wald test - joint coefficient test

lag1 vs lag3 specification comparison with sample size and number of matches maybe?

4.2 RQ1: Musculoskeletal rehabilitation

4.3 RQ2: Mortality

5 Discussion

6 Conclusion

7 Appendix

7.1 Subsection 1